

## HORBY, Sweden

WMO: 026230

Lat: 55.8633N

Lon: 13.6662E

Elev: 374

StdP: 14.50

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 11.9       | 16.4 | 7.2                           | 8.1 | 15.3 | 11.5 | 10.0 | 19.5 | 19.5                  | 41.1 | 18.0 | 40.6 | 3.6                   | 30   | 0.552 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.1                   | 79.7            | 65.4 | 76.5 | 64.0 | 73.7 | 62.8 | 67.8                | 75.8 | 65.9 | 73.0 | 64.3 | 71.2 | 6.8                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.0                            | 93.6 | 71.0 | 63.2 | 88.0 | 69.1 | 61.5 | 82.7 | 67.5 | 32.4          | 75.8 | 30.9 | 73.2 | 29.6 | 71.2 | 73.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 16.3              | 14.3 | 12.8 | 5.3                        | 83.9 | 6.6                | 3.8 | 0.6                                                | 86.6 | -3.3       | 88.8 | -6.9       | 90.9 | -11.7      | 93.7 |
|                   |      |      | 5.1                        | 71.0 | 6.5                | 1.9 | 0.4                                                | 72.4 | -3.4       | 73.6 | -7.0       | 74.7 | -11.7      | 76.1 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 46.6   | 31.2  | 31.7  | 36.1  | 44.5  | 52.8  | 58.9  | 63.1  | 62.3  | 55.6  | 47.6  | 40.2  | 34.6 |
|                                                                                  | DBStd       | 12.91  | 7.48  | 6.74  | 6.75  | 6.06  | 5.97  | 5.03  | 4.92  | 4.61  | 5.03  | 6.26  | 6.42  | 7.55 |
|                                                                                  | HDD50       | 2648   | 583   | 511   | 431   | 184   | 36    | 1     | 0     | 0     | 10    | 120   | 296   | 476  |
|                                                                                  | HDD65       | 6783   | 1048  | 931   | 895   | 615   | 381   | 195   | 97    | 110   | 284   | 541   | 745   | 941  |
|                                                                                  | CDD50       | 1423   | 0     | 0     | 1     | 19    | 122   | 267   | 407   | 382   | 179   | 44    | 2     | 0    |
|                                                                                  | CDD65       | 83     | 0     | 0     | 0     | 0     | 2     | 11    | 40    | 27    | 3     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 591    | 0     | 0     | 0     | 0     | 16    | 115   | 316   | 135   | 9     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 79     | 0     | 0     | 0     | 0     | 0     | 14    | 49    | 16    | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 6.2    | 6.9   | 6.9   | 6.8   | 6.8   | 6.4   | 5.8   | 5.3   | 5.4   | 5.7   | 6.0   | 6.4   | 6.6  |
| Precipitation                                                                    | PrecAvg     | 30.0   | 2.6   | 2.1   | 1.8   | 1.6   | 2.0   | 2.6   | 2.9   | 3.0   | 2.8   | 2.9   | 2.9   | 2.9  |
|                                                                                  | PrecMax     | 37.6   | 5.5   | 5.1   | 3.9   | 2.9   | 4.8   | 5.4   | 8.7   | 8.2   | 7.2   | 6.4   | 5.1   | 6.5  |
|                                                                                  | PrecMin     | 22.6   | 0.2   | 0.3   | 0.4   | 0.4   | 0.9   | 0.0   | 0.2   | 0.6   | 0.9   | 1.1   | 0.9   | 1.1  |
|                                                                                  | PrecStd     | 4.2    | 1.4   | 0.9   | 0.9   | 0.8   | 0.9   | 1.3   | 2.0   | 1.9   | 1.5   | 1.2   | 1.1   | 1.3  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 47.9  | 48.2  | 56.7  | 68.4  | 76.0  | 81.7  | 84.9  | 82.2  | 75.2  | 64.4  | 54.2  | 49.4 |
|                                                                                  |             | MCWB   | 46.8  | 44.3  | 48.6  | 54.9  | 61.3  | 64.2  | 67.8  | 67.5  | 64.2  | 58.8  | 53.1  | 48.7 |
|                                                                                  | 2%          | DB     | 44.5  | 45.0  | 52.2  | 63.4  | 72.3  | 77.4  | 80.7  | 77.7  | 70.5  | 60.2  | 52.0  | 47.6 |
|                                                                                  |             | MCWB   | 43.4  | 43.0  | 45.4  | 51.3  | 58.8  | 63.3  | 66.1  | 65.0  | 61.8  | 56.3  | 50.6  | 46.9 |
|                                                                                  | 5%          | DB     | 42.8  | 42.8  | 48.1  | 59.3  | 68.6  | 73.3  | 77.6  | 74.3  | 66.8  | 57.7  | 50.0  | 45.5 |
|                                                                                  |             | MCWB   | 41.7  | 41.1  | 43.4  | 49.2  | 56.9  | 61.1  | 64.9  | 63.8  | 59.2  | 55.0  | 48.7  | 44.6 |
|                                                                                  | 10%         | DB     | 40.9  | 40.8  | 45.1  | 55.3  | 64.5  | 69.8  | 74.4  | 71.3  | 63.9  | 55.7  | 48.3  | 43.6 |
|                                                                                  |             | MCWB   | 39.6  | 39.3  | 41.8  | 47.2  | 54.9  | 59.0  | 63.8  | 62.4  | 57.5  | 53.3  | 47.0  | 42.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 47.0  | 45.6  | 50.2  | 56.0  | 64.4  | 68.2  | 70.5  | 69.8  | 65.5  | 59.8  | 53.7  | 49.1 |
|                                                                                  |             | MCDB   | 47.7  | 46.9  | 54.4  | 66.2  | 73.9  | 77.9  | 79.8  | 78.2  | 72.6  | 62.8  | 53.9  | 49.3 |
|                                                                                  | 2%          | WB     | 43.7  | 43.3  | 46.5  | 52.6  | 61.0  | 64.6  | 68.4  | 67.1  | 62.6  | 57.2  | 50.9  | 47.1 |
|                                                                                  |             | MCDB   | 44.3  | 44.5  | 50.6  | 61.1  | 68.8  | 74.0  | 77.1  | 73.3  | 69.2  | 59.3  | 51.6  | 47.5 |
|                                                                                  | 5%          | WB     | 41.8  | 41.4  | 44.2  | 50.3  | 58.3  | 62.5  | 66.6  | 65.3  | 60.4  | 55.4  | 48.9  | 44.8 |
|                                                                                  |             | MCDB   | 42.6  | 42.6  | 47.2  | 56.8  | 65.3  | 71.3  | 74.6  | 71.0  | 65.1  | 57.5  | 49.7  | 45.2 |
|                                                                                  | 10%         | WB     | 39.7  | 39.3  | 42.2  | 48.0  | 55.9  | 60.4  | 64.8  | 63.8  | 58.6  | 53.6  | 47.2  | 42.8 |
|                                                                                  |             | MCDB   | 40.7  | 40.8  | 44.6  | 54.1  | 62.9  | 67.4  | 72.3  | 69.2  | 62.5  | 55.4  | 48.1  | 43.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.3   | 7.8   | 11.8  | 16.0  | 16.9  | 16.4  | 16.1  | 15.0  | 14.0  | 10.5  | 7.5   | 6.9  |
|                                                                                  |             | MCDBR  | 6.9   | 9.0   | 16.6  | 21.7  | 22.6  | 23.1  | 22.3  | 19.0  | 18.0  | 11.8  | 7.4   | 7.0  |
|                                                                                  |             | MCWBR  | 7.0   | 7.3   | 11.4  | 12.9  | 12.6  | 11.9  | 11.2  | 9.0   | 10.3  | 9.2   | 6.9   | 7.0  |
|                                                                                  | 5% WB       | MCDBR  | 6.9   | 8.1   | 14.5  | 19.2  | 19.4  | 19.9  | 18.9  | 16.6  | 16.3  | 11.4  | 7.0   | 7.2  |
|                                                                                  |             | MCWBR  | 7.2   | 7.0   | 10.9  | 12.5  | 11.9  | 11.6  | 11.0  | 9.6   | 10.3  | 9.3   | 6.9   | 7.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.305  | 0.330 | 0.352 | 0.387 | 0.384 | 0.377 | 0.399 | 0.400 | 0.368 | 0.352 | 0.332 | 0.298 |      |
|                                                                                  | taud        | 2.270  | 2.296 | 2.308 | 2.285 | 2.340 | 2.383 | 2.342 | 2.347 | 2.420 | 2.397 | 2.327 | 2.251 |      |
|                                                                                  | Ebn at Noon | 189    | 232   | 256   | 263   | 272   | 275   | 266   | 257   | 250   | 221   | 178   | 161   |      |
|                                                                                  | Edh at Noon | 17     | 24    | 31    | 36    | 36    | 35    | 36    | 34    | 28    | 22    | 16    | 14    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 137    | 328   | 767   | 1311  | 1699  | 1804  | 1669  | 1343  | 947   | 459   | 175   | 93    |      |
|                                                                                  | RadStd      | 26     | 53    | 82    | 161   | 167   | 139   | 200   | 104   | 98    | 69    | 31    | 10    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (59 neighbors) | +0.74 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -299  | +112  | N/A   |

Nomenclature: See separate page